Homework 3: 09/15 Content

MATH 1271, Garrett Credi

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Problems listed as AC are found in the Active Calculus textbook at the end of the section listed.

Problems listed as APEX are found in the APEX Calculus Textbook again at the end of the listed section.

Problems listed as PAULS-PRACT are found in Paul's Online Math Notes - Review Problems while problems listed as PAULS-ASSIGN are found in Paul's Online Math Notes - Assignment Problems and are indexed via [Topic]-[Section]-[Problem Number]

1 Type 1 - Continuity Graphically

Choose 5 of the following problems to solve.

1. AC 1.7.3
2. AC 1.7.4
3. APEX 1.5.11-18 (each of these count as one problem)
4. PAULS-ASSIGN Limits.Continuity.1-3 (each of these counts as one problem)

2 Type 2 - Continuity Algebraically

Choose 3 of the following problems to solve.

1. AC 1.7.5

2. APEX 1.5.19-34 (each of these counts as one problem)

3. PAULS-ASSIGN Limits.Continuity.4-22 (each of these counts as one problem)

3 Type 3 - One Sided Limits Graphically

Choose 5 of the following problems to solve.

1. AC 1.7.1

2. APEX 1.4.5-12

3. PAULS-ASSIGN Limits.One-Sided Limits.1-3 (each of these counts as one problem)

4 Type 4 - One Sided Limits Algebraically

Choose 4 of the following problems to solve.

1. AC 1.7.2

2. APEX 1.4.13-21 (each of these counts as one problem)

5 Type 5 - Challenge Problem

The *triangle inequality* is a result fundamental to understanding limits. It is the statement that, for real numbers a, b we always have that $|a+b| \leq |a|+|b|$.

Suppose that $f(x)$ and $g(x)$ are continuous at $x = a$, i.e. that for any output error tolerance $\epsilon > 0$ we can find an input error tolerance $\delta > 0$ so that $|x - a| < \delta$ forces $|f(x) - f(a)| < \epsilon$.

Show that $f(x)+g(x)$ is continuous at $x = a$, i.e. given some output error tolerance $\epsilon > 0$ find the right input error tolerance $\delta > 0$ so that $|x - a| < \delta$ forces $|(f(x) + g(x)) - (f(a) + g(a))| < \epsilon$.